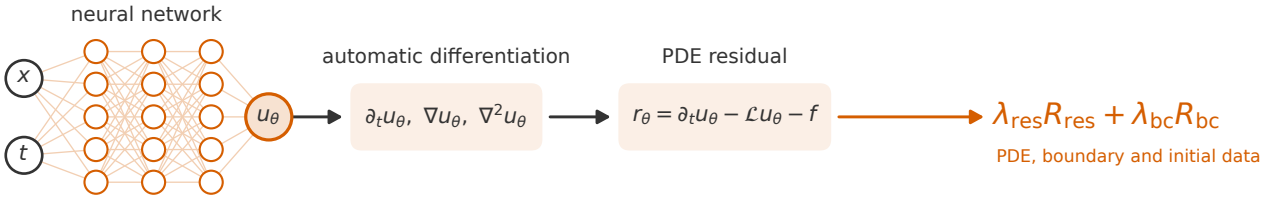
neural network
x
t
u_θ
automatic differentiation
$\partial_t u_\theta,\ \nabla u_\theta,\ \nabla^2 u_\theta$
PDE residual
$r_\theta = \partial_t u_\theta - \mathcal{L} u_\theta - f$
$\lambda_{\text{res}} R_{\text{res}} + \lambda_{\text{bc}} R_{\text{bc}}$
PDE, boundary and initial data